

Introduction to Modeling for Formal Verification

Today's goals

Module 1

Introduction to Model Checking

Motivation of Verification

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Course Updates and Accessibility Support

Reading • 1 min

●

Non-Credit Students: Welcome and Where to Find Help

Reading • 10 min

●

Motivating Examples

Video • 7 min

●

High Profile Previous Bugs and Their Impacts

Video • 10 min

●

Principles of Model Checking - Chapter 1, Introduction - Part 1

Reading • 15 min

●

Hardware and Software Bugs - Wikipedia

Reading • 20 min

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Motivation of Verification Quiz

Graded Assignment • 10 min

Overview of Verification

●

Overview of Functional Verification

Video • 5 min

●

Techniques for Functional Verification

Video • 10 min

●

Functional Verification

Reading • 30 min

●

Principles of Model Checking - Chapter 1, Introduction - Part 2

Reading • 15 min

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Overview of Verification Quiz

Graded Assignment • 10 min

Formal Verification and Model Checking

●

Overview of Formal Verification and Model Checking

Video • 7 min

●

Elements of Model Checking

Video • 6 min

●

Challenge to Model Checking: State Space Explosion

Video • 5 min

●

Principles of Model Checking - Section 1.1, 1.2

Reading • 30 min

●

Formal Verification and Model Checking - Wikipedia

Reading • 30 min

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Formal Verification and Model Checking Quiz

Graded Assignment • 10 min

Module 2

Modeling

Module 3

Modeling System Composition

Non-Credit Students: Welcome and Where to Find Help

0:15 / 4:07

2x

Hello, students!

You are enrolled in the Non-Credit version of our course and we are excited to have you here! Did you know that this course can also be taken for academic credit as part of CU Boulder’s Master of Science in Electrical and Computer Engineering (MS-ECE) degree? Additionally, this course may also count towards one of our Graduate Certificates, all of which are offered online on the Coursera platform. With performance-based admissions and no application process, the MS-ECE is ideal for individuals with diverse undergraduate backgrounds and professional experience. [Learn more about the MS-ECE program.](#) (external link; opens in a new tab)

Graduate certificates offer in-depth skill building in targeted areas and communicate your expertise to employers. Individual courses are grouped into specializations stack toward 9-12 credit graduate certificates. Students may earn free-standing graduate certificates without pursuing the entire master’s degree. Students may earn and apply graduate certificates toward the full online MS-ECE degree. These programs are an affordable way to gain industry-recognized credentials and develop advanced skills to advance your career. [Learn more about our Graduate Certificates.](#) (external link; opens in a new tab)

This post contains important information about the difference between the non-credit and for-credit versions of courses and how to reach out for assistance. Please take the time to read through it carefully as it is important in the successful completion of this course.

For-Credit and Non-Credit Course Options

You may enroll in the for-credit version of the course for the current term at any point up until the enrollment deadline (for all deadlines, see the [CU Boulder on Coursera calendar](#)). Any work completed in the non-credit version will transfer to the for-credit version upon enrollment. While the course content and assignments for the non-credit and for-credit courses are the same, there are a few important differences between the courses in terms of support.

Non-credit

The ***non-credit*** version of the course is designed to run as a free and autonomous course for anyone seeking to further their knowledge about the course topic with little to no support.

In ***non-credit*** Course Facilitators **do not**:

- Provide content help
- Provide assignment feedback
- Grade assignments
- Adjust grades
- Address discussion forum issues

For-credit

The ***for-credit*** version of the course is still autonomous, but **offers enhanced support** and access to degree planning support and additional technical assistance through the MS-ECE program support email. Additionally, **for-credit courses typically require a final project and/or exam, which distinguishes them further from the non-credit experience.**

In ***for-credit*** Course Facilitators **do**:

- Host office hours twice a week to answer student questions
- Answer emails pertaining to course content and assistance
- Review and adjust assignment grades upon request

If you are considering the for-credit version, enrolling sooner rather than later allows you to benefit from additional support and interaction with facilitators, ensuring a more enriching learning experience.

Where to Find Help

Non-credit learners have a few options if they run into problems. We recommend using the following resources in the order listed:

Technical Support

1. Find related articles in the Learner Help Center
2. [Contact Coursera's technical support team](#) (external link; opens in a new tab)
3. [Flag the issue](#) in the course (non-urgent issues only)

Course Content Support

1. Ask your peers for support or suggestions in a discussion forum
2. Utilize Coursera Coach in available readings and videos to further your understanding of the content

We work to promptly fix all issues that Coursera technical support identifies as requiring the course author's attention. Please know that we prioritize urgent issues that impede learner progress and we also periodically review flagged issues.

Mark as completed

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